NATIONAL UNIVERSITY of Computer & Emerging Sciences

RESEARCH PROJECT

COMPUTER NETWORKS

PHASE 3 — RESEARCH IMPLEMENTATION AND EXPERIMENTAL EVALUATION

As you have completed **Phase 2: Research Methodology Design**, including your research problem, identified research gap, proposed methodology, and methodology architecture, you are now required to proceed with **Phase 3: Research Implementation and Experimental Evaluation**.

Your implementation must include the following components:

1. **Implementation of the Proposed Methodology:** Implement your proposed model, algorithm, or framework according to the methodology designed in Phase 2. Incorporate the feedback received during the previous class and document any changes made to your original methodology.
2. **Dataset and Preprocessing:** Collect or select the required dataset, perform the necessary preprocessing and feature engineering, and clearly document the experimental setup, data splits, and implementation details.
3. **Baseline Evaluation:** Implement or use appropriate existing baseline methods and evaluate their performance using the relevant metrics for your research problem. Include all essential baseline evaluation metrics.
4. **Graphical Results and Visualization:** Present your experimental results through appropriate graphs, charts, and visualizations. The figures should clearly demonstrate the performance, behavior, and effectiveness of your proposed methodology.
5. **Comparison with Existing Research:** Compare your proposed methodology with existing approaches reported in relevant research papers. Use comparable

datasets, evaluation metrics, and experimental settings wherever possible, and clearly identify any limitations affecting a fair comparison.

6. **Ablation Studies:** Conduct different ablation experiments to investigate the contribution of individual components of your proposed methodology. Remove, replace, or modify selected components and analyze how these changes affect the overall performance.
7. **Results Analysis and Discussion:** Interpret your findings, discuss the strengths and limitations of your proposed approach, and explain whether the experimental results address the research gap and objectives identified in Phases 1 and 2.

Expected Outcome: Submit a complete implementation and experimental evaluation of your proposed research methodology, supported by reproducible experimental details, baseline comparisons, graphical results, ablation studies, and a critical discussion of the findings.